STAFF REVIEW · H446 008 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q2; Booklet Q2 = original online Q5

Do Now — source answers

1. String: preserve the leading zero; it is an identifier, not a quantity for arithmetic.

2. 65; 01000001.

3. 00000101. The low four bits are preserved; the high four are cleared.

4. 3 − 2 = 1 → 010000 0001.

I Do — source worked steps

1. Start with the character codes — CODE:
A = 65
B = 66 — ASCII assigns a numeric code to each character.

2. Write the 7-bit ASCII values — CODE:
A = 1000001
B = 1000010

3. Add a leading zero — CODE:
A → 01000001
B → 01000010

AB → 01000001 01000010 — The leading zero lets each 7-bit ASCII code sit inside an 8-bit byte; it does not expand the standard ASCII repertoire.

We Do — guided CHECK

1. Build the answer

2. Build the answer
For lowercase c : 97 + 2 = 99; uppercase C = 99 − 32 = 67.

3. Build the answer
Both ends must interpret the bytes using the same encoding.

4. Build the answer

5. Build the answer

Page 3 — independent skill check, teacher-review model

1. Standard ASCII has 128 codes (7-bit), unsuitable for Chinese characters; Unicode has a far wider repertoire. C = 67 = 01000011 in an 8-bit byte.

You Do — source mark schemes

1. CODE:
Gap 1: AND
Gap 2: count + 1
Gap 3: count
One mark each. Trace check: AaZ! has two uppercase letters. OR would incorrectly accept codes outside the uppercase range.

2. ASCII lacks Chinese characters (1). Unicode includes the required writing systems (1). Both assign numeric codes to characters (1). The receiver needs the encoding to interpret the byte sequences as the intended characters (1).

3. CODE:
C: 65 + 2 = 67 → 01000011
c: 97 + 2 = 99 → 01100011
'5': 48 + 5 = 53 → 00110101
For each: code calculation (1), correct byte (1).

4. 128 (1). Eight bits provide 256 codes, which is insufficient (1); nine bits provide 512, so the minimum is 9 (1).

5. CODE:
(20 × 1) + (3 × 3) + (2 × 4) = 37 bytes
37 × 8 = 296 bits
Correct weighted calculation (1), 37 bytes (1), 296 bits (1). UTF-8 has variable byte widths; the supplied groups do not all use two bytes (1).

Exit Ticket — source answer

1. True. Standard ASCII has 7 bits and therefore 128 possible codes.